

Dataset and benchmark limitations

Project: Reproducible AI Research Study

1. Recall and choose your starting point

Try the recall before opening notes. Then use a different colour or a labelled second line to correct your explanation.

Sensitivity analysis

Without opening the old lesson, explain one key idea from Sensitivity analysis. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

Multiple comparisons

Without opening the old lesson, explain one key idea from Multiple comparisons. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

Uncertainty intervals

Without opening the old lesson, explain one key idea from Uncertainty intervals. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

2. Practise with fading help

Use W2: Trace the worked case for Dataset and benchmark limitations. Record the answer once; the lesson and workbook refer to the same attempt.

Given inputs · Procedure I chose · Middle steps · Result and check

Given inputs: _____

Procedure I chose: _____

Middle steps: _____

Result and check: _____

The step I completed without help, and the hint I needed

3. Individual reasoning check

Use the worked case for Dataset and benchmark limitations. Proposed explanation: “The worked result establishes every future case”. Decide whether this explanation is supported. Point to the step or supplied fact that decides; explain your repair if needed.

My explanation before partner discussion

4. Independent transfer case

Independent case: Apply a new case independently

Twelve rows include four copies of one source and eight others. Count unique units and name one coverage concern.

My independent explanation, trace and limits

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

5. W1 Explain the mechanism

Explain the weekly mechanism in your own words. Use the worked scenario to name its input, decision rule and output.

My reasoning and evidence

6. W2 Trace the worked case

Ten benchmark rows include three copies of one source and seven other unique sources. Show the intermediate steps and give the conclusion with its scope.

My reasoning and evidence

7. W3 Audit the limit

A report omits this qualification: Benchmark success does not establish performance on a deployment population missing from its design. Explain how omitting it could change a reader's interpretation, and write an accurate replacement.

My reasoning and evidence

8. W4 Apply a new case independently

Twelve rows include four copies of one source and eight others. Count unique units and name one coverage concern.

My reasoning and evidence

9. W5 Design a revealing follow-up

For the worked scenario, propose one additional case that could reveal a failure, describe its expected outcome before running it, and explain what would need to remain fixed for a fair comparison. Do not reuse W4 as a new final test after receiving feedback.

My reasoning and evidence

10. Project checkpoint and reflection

Add a labelled dataset and benchmark limitations evidence sheet to Reproducible AI Research Study. Show how you can explain the mechanism and terminology of dataset and benchmark limitations. Use the supplied fictional case and keep its inputs, intermediate steps, calculation or prediction, and limitation. Connect this record to last week's record; a matching answer without a trace is insufficient.

My project change, supporting evidence and remaining limit

11. Feedback, return check and learning record

Evidence · My record

Evidence: Worked alone / hint / model

My record: _____

Evidence: First step I repaired and why

My record: _____

Evidence: New case checked later; date and result

My record: _____

Evidence: What I can now explain without notes

My record: _____

Evidence: One remaining question

My record: _____

Revisit this idea with your teacher after a delay. Familiarity is different from explaining and using it on a changed case.

12. Pictorial case practice: Dataset and benchmark limitations

A benchmark can miss the intended setting. A plant model is evaluated on clean indoor photos but proposed for shaded outdoor photos.

Indoor benchmark: 9/10 correct. No outdoor cases are present. The same model is intended for both.

Benchmark conditions

	Correct	Total
Indoor benchmark	9	10
Added outdoor slice	2	4

Read a cell using both its row and column labels.

Dataset and benchmark limitations

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.



1. Name measured setting

All ten benchmark images are clean indoor photos.



2. Report the result

The observed indoor score is 90%.



3. Name missing evidence

Outdoor shade performance remains unmeasured.

CHANGE ONE THING

Add four reserved outdoor cases, of which two are correct.

Explain the changed result using the matching lesson.

Report the indoor benchmark with its denominator and missing condition. After two correct outdoor cases out of four are added, what slice rate is supported and what remains uncertain?

My picture-based explanation and changed-case prediction

Support used: alone / hint / worked example. This record is practice evidence.

Project connection: For the Reproducible AI Research Study, retain an annotated worked trace and the changed-case reasoning. Connect the claim to its records, units and assumptions so another investigator can challenge or reproduce it.

13. Project portfolio milestone

Project: Reproducible AI Research Study

Current milestone: Test changed cases. Keep normal, boundary and failure cases; compare expected and observed results.

For the Reproducible AI Research Study, retain an annotated worked trace and the changed-case reasoning. Connect the claim to its records, units and assumptions so another investigator can challenge or reproduce it.

Evidence to keep · My record

Evidence to keep: This week's input and deciding step

My record: _____

Evidence to keep: Prediction and actual result (or labelled paper trace)

My record: _____

Evidence to keep: One change since the previous checkpoint and why

My record: _____

Evidence to keep: One error or limit and the next test

My record: _____

Explain what this week adds to your project. Reuse last week's record rather than starting again.

14. Supported practice and learning evidence

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

A benchmark can miss the intended setting. A plant model is evaluated on clean indoor photos but proposed for shaded outdoor photos.

Practice 1: explain the deciding evidence

Which labelled step matches this detail? Outdoor shade performance remains unmeasured.

1. Report the result
2. Name measured setting
3. Name missing evidence

My selected conclusion, deciding fact and explanation

Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

After your attempt, compare with the supported explanation: Name missing evidence Outdoor shade performance remains unmeasured. A benchmark represents particular tasks and collection conditions. High performance on it cannot automatically establish fitness for a different deployment setting.

Practice 2: explain the deciding evidence

Changed-case practice: Add four reserved outdoor cases, of which two are correct. Which conclusion is supported by the supplied facts?

1. Report the outdoor slice as $2/4=50\%$, separately from indoor results.
2. We can decide the result without examining the changed input or the procedure.
3. This one example guarantees correct results for every future input.

My selected conclusion, deciding fact and explanation

Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

After your attempt, compare with the supported explanation: Report the outdoor slice as $2/4=50\%$, separately from indoor results. The new records finally measure the missing condition, with a small denominator.

Repair a missing building block

Review Sensitivity analysis (ai-7/17). Goal: Explain the mechanism and terminology of sensitivity analysis. Short practice: Classify 0.4, 0.6 and 0.8 at cutoffs 0.5, 0.7 and exactly 0.6. Which equality rule decides the boundary, and can these unlabelled scores identify the best cutoff?. Return to this week and explain what you changed.

Review Multiple comparisons (ai-7/15). Goal: Explain the mechanism and terminology of multiple comparisons. Short practice: Calculate the six-test cutoff and identify which supplied p-values reject. Why must the team count every planned comparison rather than report only its smallest p-value?. Return to this week and explain what you changed.

Review Uncertainty intervals (ai-7/12). Goal: Explain the mechanism and terminology of uncertainty intervals. Short practice: Locate -0.8 and zero relative to the supplied interval $[-1.4, -0.2]$. Interpret the sign and confidence-method meaning, then describe what changes with interval $[-1.4, 0.3]$.. Return to this week and explain what you changed.

Keep your completed work

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run evidence. Open the HTML copy to read or print it. Drafts stay in this browser when storage is available; clear drafts on a shared device.

What I repaired, the evidence I kept and the next test

